Multiscale Sparse Cross-Attention Network for Remote Sensing Scene Classification

Jingjing Ma, *Member, IEEE*, Wei Jiang, Xu Tang, *Senior Member, IEEE*, Xiangrong Zhang, *Senior Member, IEEE*, Fang Liu, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Remote sensing (RS) scene classification (RSSC) is a prominent research topic in the RS community. Multilevel feature fusion is an important way of addressing RSSC, and many methods have been proposed in recent years. Although they succeed, current methods can still be improved, particularly in distinguishing the contributions of different multilevel features and fully and effectively fusing them. To address the above issues and fully exploit the potential of multilevel features for RSSC tasks, we propose a new model named multiscale sparse cross-attention network (MSCN). It not only focuses on the effectiveness of feature learning but also emphasizes the rationality of feature fusion. In detail, MSCN first extracts multilevel features using a pretrained ResNet50. Also, these features are divided into high- and low-level features according to the clues they are involved. Then, a multiscale sparse cross-attention (MSC) module is developed to cross-fuse the high-level feature with various low-level features, thereby effectively mining helpful information from multilevel features. In the fusion process, MSC not only explores the multiscale messages in RS scenes but also mitigates the negative impact of irrelevant information by employing sparse operations. Third, a group convolutional block attention module (CBAM) enhancer (GCE) is presented to enhance the representation of classification features. GCE detects local salient information within classification features using grouped CBAM and further enhances crucial details by readjusting the CBAM attention weights. This way, the classification features' discrimination can be improved. We conducted extensive experiments on three public RSSC datasets. The exceptional experimental results indicate that our proposed MSCN achieves superior classification accuracy, surpassing many existing methods. Our source codes are available at https://github.com/TangXu-Group/Remote-Sensing-Images-Classification/tree/main/MSCN.**

***Index Terms*— Feature enhancement, remote sensing scene classification (RSSC), sparse cross-attention.**

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Jingjing Ma, Wei Jiang, Xu Tang, Xiangrong Zhang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: tangxu128@gmail.com).

Fang Liu is with the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210000, China.

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I. Introduction

REMOTE sensing (RS) scene classification (RSSC) refers to assigning the specific semantics to RS images according to their scene-level contents. It is an important research topic in the RS image processing community, and it plays a crucial role in various applications, including land use investigation, environmental protection, and so on [1], [2], [3]. In the early stages, handcrafted features, such as geometry textural [4] and color histogram [5], are popular in RSSC due to their simple implementation. Along with the machine-learning-based classifiers, acceptable results can be achieved. However, with the improvement in spatial resolution, the contents of RS scenes are more and more complex. The combination of handcrafted features and traditional classifiers is becoming increasingly inadequate. To obtain reliable classification predictions, the deep learning models, especially convolutional neural networks (CNNs) [6], [7], [8], enter the RSSC field [9], [10], [11], [12]. Benefiting from their powerful feature learning capability and end-to-end learning manner, the classification accuracy of RSSC has been elevated to a new height.

Numerous successful CNN-based methods have been proposed [13], [14], [15], [16]. They apply a series of convolution and downsampling operations to RS scenes to extract high-level features for classification. Although the CNN-based methods perform well, some issues are also introduced. For example, small-sized targets tend to disappear as the number of convolutional layers increases, which leads to information loss and interferes with the final decision. To further illustrate this problem, we adopt ResNet50 [17] to extract an RS scene's features and visualize them orderly in Fig. 1. Here, the RS scene is randomly selected from the UC Merced land use [18] dataset, and it contains planes with various sizes. The chosen RS scene and visual feature maps (from low to high level) are exhibited in the figure below. We can easily find that the five small planes in the middle and bottom parts gradually disappear as the convolutional layers increase. When the high-level feature map is used, the generated classification result may be incorrect. To address the above trouble, the direct and effective solution is to leverage multilevel feature information simultaneously. In other words, exploring the multiscale knowledge hidden in RS scenes is important for classification, which is also our crucial motivation. One common approach is to employ dense connections [19] to

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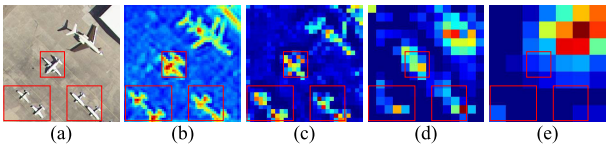


Fig. 1. RS scene and its visual feature maps extracted by ResNet50. The RS scene is selected from the UC Merced land use dataset, and it contains planes with various sizes. (a) RS scene. (b)–(e) Visual feature maps output by the first, second, third, and fourth residual blocks.

preserve information from each stage of the CNN. However, densely connected networks require substantial computational and memory resources, limiting the models' generalization ability in practical tasks [20]. Another way involves hierarchical feature fusion [21], which focuses on effectively merging the extracted hierarchical features. Common fusion techniques include the feature addition [22], multiplication [23], and channel splicing [24]. Although these methods perform well, they do not consider the redundant information and the intricate relationships among these multilevel features [25]. Recently, transformer-based models like vision transformer (ViT) [26] and generative pretrained transformer (GPT) [27] models have emerged and achieved significant success in RSSC [28], [29]. The key of transformer is the self-attention mechanism (SAM), which focuses on exploring the global relationships within the input data [30]. When the input data covers different sources, SAM would be transformed into the cross-attention mechanism (CAM) [30] smoothly, and many works use it to fuse multilevel features to generate discriminative feature representations [31], [32], [33]. With the help of these features, satisfactory classification results can be produced.

The above-mentioned RSSC works have achieved impressive results; however, there is still room for improvement. First, the multilevel features extracted by CNNs contain diverse clues. High-level features are rich in semantics, while low-level features contain much detailed knowledge, such as texture and edges [34]. From the perspective of the nature of the clues they contain, these two features complement each other. However, many methods use them directly without considering their contributions, which leads to an incomplete understanding of the contents across different scales in RS images. Second, fusing the low- and high-level features with the specific attention mechanism [30] is a common and valuable solution to enhance the visual representation for RSSC tasks. Nevertheless, since the contents within RS images are complex, the details within low-level features would be huge in volume and diverse in type. Among these details, not all of the knowledge is beneficial for RSSC tasks. Therefore, it is important to identify useful information and discard interference items during the feature fusion to improve the results of RSSC. Finally, after the feature fusion, there is always a feature enhancement step for the final classification features. Current methods [35], [36] usually adopt the feature analysis tool to achieve this from the global aspect, such as convolutional block attention module (CBAM) [37]. However, they may disrupt the representation of information among different local channels, limiting the classification features' discrimination.

This article develops an RSSC model named multiscale sparse cross-attention network (MSCN) to address the above problems. First, MSCN utilizes a pretrained ResNet50 network [17] to extract multilevel features from RS images. The outputs of the first three blocks within ResNet50 are treated as shallow features, and the output of the last block is regarded as a high-level feature. Considering that the contributions of various shallow features are different, instead of fusing them into the high-level feature together, we combine them with the high-level feature individually. This way, the valuable clues hidden in the multilevel features can be highlighted for the classification. Second, rather than adopting SAM or CAM directly, a multiscale sparse cross-attention (MSC) mechanism is proposed for multilevel feature fusion. On the one hand, MSC can explore complex content with various scales hidden in RS scenes. On the other hand, it typically ignores irrelevant information by the sparse scheme, reducing the impact of interference items. Third, a group CBAM enhancer (GCE) is introduced to enhance the feature representations. GCE divides the fused feature into several parts along the channel dimension and applies CBAM to each part to find the salient messages locally. Then, all local clues are concatenated to generate the classification feature. This local-to-global enhancement scheme ensures the discrimination of obtained representation.

The main contributions of this article are as follows.

1) We propose an RSSC network named MSCN. It considers the varying contributions of multilevel features extracted by CNN and cross-integrates the high-level feature with each shallow feature. This approach effectively mines the useful information embedded within the multilevel features.
2) MSC is introduced to fuse multilevel features effectively. On the one hand, it focuses on exploring complex content at various scales hidden in RS scenes. On the other hand, it typically reduces the impact of irrelevant information through a sparse scheme, thereby minimizing the influence of interfering items.
3) GCE is developed to enhance the representation of the classification feature. It identifies local salient information within the classification feature by employing a group of CBAM. Additionally, GCE enhances key information by readjusting CBAM attention weights, making the classification feature more discriminative.
4) Comprehensive experiments are conducted on three datasets, and the positive results indicate the effectiveness of MSCN in RSSC tasks.

The rest of this article is summarized as follows. The published RSSC works on multilevel feature fusion and sparse representation in SAM are reviewed in Section II. Section III describes the proposed MSCN. We present the experimental results and analysis in Section IV. At last, a short conclusion is drawn in Section V.

II. Related Work

A. Background of RSSC

RSSC involves interpreting the content of a given RS image and assigning a corresponding semantic label to the

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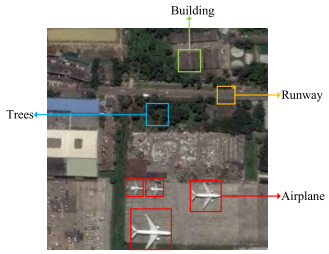


Fig. 2. RS scene selected from Northwestern Polytechnical University-RS image scene classification 45 (NWPU) dataset. Its semantic label is Airplane. There are four planes in this picture, and their scales are varied. Meanwhile, besides the airplanes, many other land covers exist, such as trees, building, and runway, which are considered "irrelevant" in single-label classification tasks. Thus, capturing multiscale clues and depressing the "irrelevant" contents are important to RSSC.

identified scene [38]. The key challenge in this task lies in feature extraction, which typically includes texture, spectral, and semantic features [39]. Unlike natural scene classification, RS scenes often encompass many land covers, and they are diverse in type and scale, thereby complicating the classification process [40]. Although the models [13], [14], [15], [16] developed based on CNN, ViT [41], and Mamba [42], [43], [44] achieve remarkable results in RSSC tasks, further improving classification accuracy requires a deeper consideration of the unique characteristics of RS images.

First, the scale clues hidden in RS scenes are abundant. Taking Fig. 2 as an example, which is chosen from the Northwestern Polytechnical University-RS image scene classification 45 (NWPU) dataset [38], its semantic label is Airplane. There are four airplanes, and their sizes are varied dramatically. Ensuring that information from larger planes can be captured entirely and smaller planes would not be lost during feature learning is a key challenge when using deep networks for RSSC tasks. Second, the single labels cannot cover the generous contents within RS scenes. For instance, although the semantic label of the RS scene in Fig. 2 is Airplane, many other land covers exist, such as trees, building, and runway. These "irrelevant" contents would disturb the deep networks, leading to misclassification. Based on the discussion above, a useful RSSC model should be capable of capturing multiscale messages from the RS image, effectively integrating them, and suppressing the influence of "irrelevant" information. These are precisely the key issues that the proposed MSCN aims to address.

B. Multilevel Feature Fusion for RSSC

Multilevel feature fusion is a helpful technique for RSSC, in which the features from different scales are integrated subtly to describe the complex contents hidden in RS images. With the assistance of these intact representations, good RSSC results can be achieved. In this section, we divide the popular methods into two groups for review, i.e., feature-combination- and CAM-based models.

Feature-combination-based fusion methods combine different features according to some specific rules, such as concatenation, addition, and multiplication. For example, Bi et al. [19] proposed a multiple instance densely connected convolution network (MIDC-Net) for RSSC. MIDC-Net first develops a densely connected CNN to learn low- and mid-level features from RS images. The proposed network simplifies the structure of the dense net [45] to reduce the computational load and effectively preserve the shallow and middle features. Then, various features are fused by the simple concatenation for classification. The reported results confirm the usefulness of the densely connected CNN and the feasibility of multilevel feature fusion. However, the fusion scheme is too simple, so the final performance cannot meet what we expected. To solve this problem, another feature-combination-based model was introduced in [41] named spatial–frequency multiscale transformer (SF-MSFormer). It extracts spatial features through the feature pyramid network (FPN) [46] and spatial transformer. In addition, it utilizes the multilevel adaptive wavelet transform and frequency multiscale transformer to capture global frequency texture features. To facilitate effective fusion, SF-MSFormer employs adaptive convolutional operations to adjust their sizes before concatenating them along the channel dimension. A similar work was published in [23], in which a feature correlation enhancement module is first developed to discover the relations between features from different levels. Then, a channel splicing strategy is proposed to combine these features. In addition, considering the spatial differences in the fused features, a spatial refinement module is introduced to reduce differences between shallow and deep features further. The encouraging results demonstrate the work's effectiveness. To deeply investigate the relationships among the multilevel features for fusion, Chen et al. [32] proposed a hierarchical feature fusion of transformer with patch dilating (HFFT-PD). By whittling down the semantic gaps between different features and ensuring the important channels in various features, the multilevel features can be integrated completely, enhancing the final classification performance. Chen et al. [42] proposed an RSSC model based on Mamba, named RSMamba. It utilizes three scanning strategies to model the global relationships in RS scenes. Subsequently, the features obtained from different path scanning strategies are fused by simple additions for the final classification. RSMamba overcomes the limitation of the Mamba in spatial position perception, and its experimental results demonstrate its effectiveness.

Compared to these methods, MSCN accounts for the varying contributions of the local features at different levels. MSCN adopts a progressive feature fusion strategy rather than combining multilevel features directly, incrementally extracting complementary information from features at different levels, which effectively focuses on targets that are prone to being lost in deeper layers, thereby enriching the feature representation. Furthermore, compared to simple feature-combination-based fusion strategies, MSCN utilizes the attention mechanism to fuse features, enabling it to model long-range dependencies between different features. It dynamically selects key information for fusion, generating more discriminative and representative fused features.

CAM-based fusion methods integrate the features under the paradigm of SAM. The different features are first mapped into the Keys, Queries, and Values. Then, they are fused according

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to the SAM's rule. Compared to the feature-combination-based methods, the models in this group can globally explore the relations within different features, enhancing the capacity of fused representations. Chen et al. [47] introduced a two-branch network called CrossViT to investigate the effectiveness of multiscale features within ViT networks. Each branch of CrossViT processes image patches of varying sizes. Leveraging the CAM, CrossViT selectively fuses information from one branch only with the classification token from another branch, thereby reducing computational requirements while ensuring sufficient information interaction. The favorable results obtained with CrossViT serve as evidence of its effectiveness. Another CAM-based model was introduced in [48] named efficient multiscale transformer and cross-level attention learning (EMTCAL). On the one hand, EMTCAL adopts ResNet [17] to extract multilevel features from the local aspect. On the other hand, EMTCAL designs an efficient ViT to mine contextual information hidden in these features. Finally, the features across various levels are fused by a CAM. The positive results illustrate that EMTCAL is helpful in RSSC tasks. Similarly, Yuan and Wang [49] proposed an RSSC network that integrates CNN and ViT, which is called a hierarchical contextual feature-preserved network (HCFPN). First, HCFPN extracts multilevel features from RS scenes using CNN. Then, it captures contextual features from the first two multilevel features using ViT. Finally, HCFPN leverages CAM to combine different clues with high-level semantic features for classification. The promising results confirm it is helpful for RSSC.

Unlike these methods, MSCN improves upon the CAM to suit RS scenes better. Specifically, MSCN extracts multiscale information through pooling operations at different scales, enabling the capture of the abundant scale clues hidden in RS scenes. Additionally, MSCN employs a sparse fusion strategy to select more critical information from local features, effectively reducing the negative impact of "irrelevant" clues detrimental to single-label classification tasks.

C. Sparse Representation in SAM

SAM typically requires computing attention weights for all sequences to analyze the input data. This is a time-consuming process, and it would be influenced by the interference information from less correlated sequences, thereby limiting its applicability. To mitigate these challenges, researchers try introducing the idea of sparse representation [50] into the attention mechanism. For example, Liu et al. [51] proposed the shifted window (SWin) Transformer, which restricts the attention mechanism to local windows and shifts the window at different stages to ensure understanding of global contextual relationships. Compared to full attention, the SWin Transformer significantly improves computational efficiency while ensuring a comprehensive understanding of image information. Inspired by the SWin Transformer, a sparse attention network called cross-shaped window (CSWin) Transformer was proposed in [52], which performs local attention calculations on the input features within horizontal and vertical stripes. It enables the model to focus on important features in the intersection area while reducing the computational complexity of full attention, maintaining a balance between global and local information. The excellent results indicate that the CSWin Transformer performs better with reduced computational load. Likewise, Zhu et al. [53] introduced a sparse network called bi-level transformer (BiFormer), which innovatively employs dual-path attention. Specifically, BiFormer initially conducts attention calculation at a coarser granularity to filter out irrelevant key-value pairs for each query. Then, token-to-token attention is applied to these filtered key-value pairs. Compared to full-attention networks, BiFormer has a smaller computational load and reduces the impact of interference information in the image. The outstanding performance exhibit its effectiveness. To improve the downstream tasks efficiency, Ye et al. [54] presented a candidate elimination mechanism in ViT. This mechanism preserves the attention scores between templates and patches in the search region, subsequently discarding search region patches with low scores at each stage. As a result, the computational burden is reduced, and interference from irrelevant search region patches is minimized.

III. Proposed Method

The overall framework of MSCN is depicted in Fig. 3. It comprises four main components: a multilevel feature extraction module (MFEM), a multilevel feature cross-fusion module (MFCM), a GCE, and a classification module (CM). MFEM aims to learn multilevel features from RS images. MFCM is developed to fuse the high-level feature with different low-level features to obtain classification features. GCE is employed to enhance the classification features further. CM completes the final classification. Now, we introduce them in detail.

A. Multilevel Feature Extraction Module

MFEM first leverages ResNet50, a widely used CNN model in the RS community, to extract the multilevel features from RS images. ResNet50 has four residual blocks, each of which outputs a feature map corresponding to a certain level. For convenience, we record these multilevel outputs $\mathbf{F}_1 \in \mathbb{R}^{H_1\times W_1\times C_1}$, $\mathbf{F}_2 \in \mathbb{R}^{H_2\times W_2\times C_2}$, $\mathbf{F}_3 \in \mathbb{R}^{H_3\times W_3\times C_3}$, and $\mathbf{F}_4 \in \mathbb{R}^{H_4\times W_4\times C_4}$, where H_i, W_i, and C_i stand for the height, width, and channel of $\mathbf{F}_i$. From $\mathbf{F}_1$ to $\mathbf{F}_4$, features undergo a gradual transformation from low to high level, with their content shifting gradually from texture details to advanced semantic information.

Then, MFEM employs convolution operations on $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ to adjust their channels to C_2 and reduce the size of them to their half. Reducing their size aims to reduce the computational costs in the subsequent fusion stage, and utilizing 3×3 convolution for size reduction helps to retain more useful feature information. This progress can be formulated as follows:

$$\mathbf{F}'_i = \mathrm{Conv}_{3\times 3}(\mathbf{F}_i), i = 1, 2, 3 \tag{1}$$

where $\mathrm{Conv}_{3\times 3}(\cdot)$ indicates the 3×3 convolution with a stride of 2 and padding of 1. For $\mathbf{F}_4$, we only adjust its channels to C_2 by a 1×1 convolution, which is defined as follows:

$$\mathbf{F}'_4 = \mathrm{Conv}_{1\times 1}(\mathbf{F}_4) \tag{2}$$

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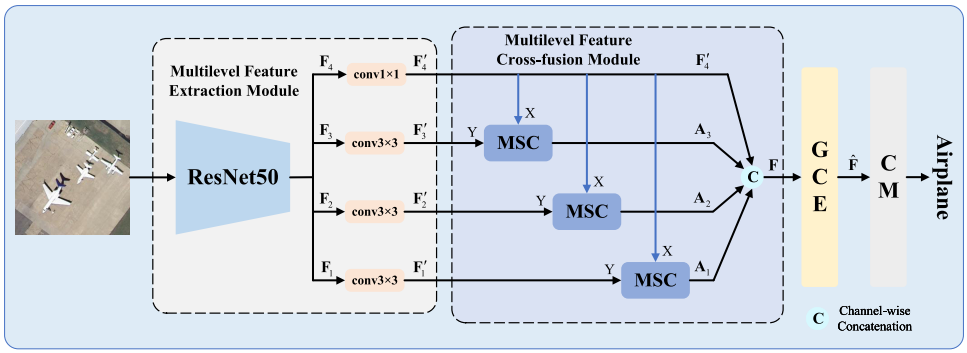


Fig. 3. Framework of MSCN. It consists of four parts: an MFEM, an MFCM, a GCE, and a CM.

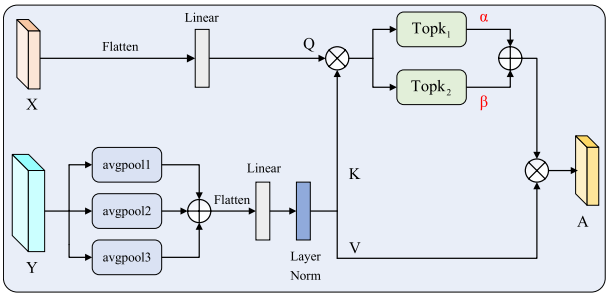


Fig. 4. Schematic of MSC. It first employs varying window sizes for average pooling to capture multiscale clues in the input data. Then, two TopK sparse operations are adopted to fuse the different features considering the irrelevant information.

where $\mathrm{Conv}_{1\times1}(\cdot)$ indicates the 1×1 convolution with a stride of 1 and padding of 0.

So far, we obtained the multilevel features $\mathbf{F}'_1 \in \mathbb{R}^{(H_1/2)\times(W_1/2)\times C_2}$, $\mathbf{F}'_2 \in \mathbb{R}^{(H_2/2)\times(W_2/2)\times C_2}$, $\mathbf{F}'_3 \in \mathbb{R}^{(H_3/2)\times(W_3/2)\times C_2}$, and $\mathbf{F}'_4 \in \mathbb{R}^{H_4\times W_4\times C_2}$ from RS images.

B. Multilevel Feature Cross-Fusion Module

In MFEM, we obtain the multilevel features $\mathbf{F}'_1$, $\mathbf{F}'_2$, $\mathbf{F}'_3$, and $\mathbf{F}'_4$. The high-level feature $\mathbf{F}'_4$ contains advanced semantic information, while the low-level features $\{\mathbf{F}'_1, \mathbf{F}'_2, \mathbf{F}'_3\}$ enclose much detailed knowledge. They complement each other. Therefore, we hope to extract information from various low-level features to enrich the representation of high-level feature, making it more comprehensive and discriminative for the final classification. However, how to fuse these features effectively is a key problem. The contents of RS images are diverse in type and varied in scale, and the land covers with different scales provide varying levels of semantic and structural details. Therefore, considering these multiscale clues during the fusion process is the first challenge. Additionally, not all information hidden in the multilevel features is useful. The interference messages can affect the fusion results negatively. Hence, identifying and eliminating them in the fusion step is another challenge. To address these challenges, we propose MSC in this section.

The schematic of MSC is shown in Fig. 4. The key idea of MSC comes from CAM [30], which combines the different inputs by using the attention mechanism. Unlike CAM, the proposed MSC not only considers multiscale clues in RS scenes but also effectively suppresses irrelevant information interference through sparse operations, resulting in more accurate feature fusion outcomes. Suppose there are two inputs $\mathbf{X} \in \mathbb{R}^{h_1\times w_1\times c}$ and $\mathbf{Y} \in \mathbb{R}^{h_2\times w_2\times c}$, where h_1 and h_2, w_1 and w_2, and c represent the heights, widths, and channels. To obtain multiscale clues hidden in the input data, MSC first applies three average pooling operations with different windows to $\mathbf{Y}$ to generate $\mathbf{Y}_1$, $\mathbf{Y}_2$, and $\mathbf{Y}_3$. Note that their shapes are consistent with $\mathbf{Y}$. The larger pooling window can capture the overall structural information in RS images, while the smaller one can capture detailed information. Then, these pooled features are added together to produce $\hat{Y}$, which contains abundant multiscale information. The details of the average pooling operations can be found in Section IV-C1. Next, $\mathbf{X}$ and $\hat{Y}$ are flattened and mapped into $\mathbf{Q} \in \mathbb{R}^{N_1\times d}$, $\mathbf{K} \in \mathbb{R}^{N_2\times d}$, and $\mathbf{V} \in \mathbb{R}^{N_2\times d}$ (N_1 equals $h_1 \times w_1$, N_2 equals $h_2 \times w_2$, and d equals c) according to the following formula:

$$\begin{aligned} \mathbf{Q} &= \mathrm{Flatten}(\mathbf{X}) \cdot \mathbf{W}_1 \\ \mathbf{K}, \mathbf{V} &= \mathrm{Split}\big(\mathrm{LN}\big(\mathrm{Flatten}\big(\hat{Y}\big) \cdot \mathbf{W}_2\big)\big) \end{aligned} \tag{3}$$

where $\mathrm{Flatten}(\cdot)$ is the flatten operation, $\mathrm{Split}(\cdot)$ is the uniformly split operation along the channel dimension, $\mathrm{LN}(\cdot)$ is layer-norm function, and $\mathbf{W}_1$ and $\mathbf{W}_2$ are linear projection parameters. Also, the attention map $\mathbf{M} \in \mathbb{R}^{N_1\times N_2}$ can be obtained by

$$\mathbf{M} = \mathbf{Q} \times \frac{\mathbf{K}^{\mathbf{T}}}{\sqrt{c}} \tag{4}$$

where c is the dimension of $\mathbf{Q}$.

Here, $\mathbf{M}$ illustrates the similarity relationships between patches in $\mathbf{Q}$ and $\mathbf{K}$, and determines how to allocate attention weights to $\mathbf{V}$. Specifically, the item m_{ij} in $\mathbf{M}$ stands for the correlation between the ith element in $\mathbf{Q}$ and the jth one in $\mathbf{K}$. The smaller the value, the lower the correlation, indicating the jth element is irrelevant. The ith row of $\mathbf{M}$ signifies the similarity between all elements in $\mathbf{K}$ and the ith item in $\mathbf{Q}$. To reduce the negative impacts of irrelevant elements, we use a TopK sparse operation [55] to abandon the lowest elements with a proportion of $1-k$ and reserve the highest k proportion in each row of $\mathbf{M}$. This progress is shown in Fig. 5. Note that we replace the irrelevant elements' values by $-\infty$ rather than removing them. This way, these values are transformed into 0 through the softmax operation. The operation discussed

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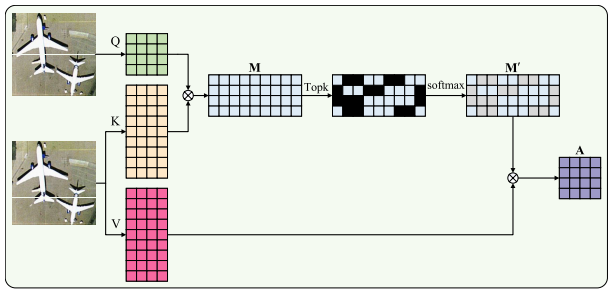


Fig. 5. Illustration of TopK sparse operation. Black color represents the value of $-\infty$, while gray indicates the value of 0.

above can be formulated as follows:

$$\mathbf{M}' = \text{Softmax}(\text{Topk}(\mathbf{M})) \tag{5}$$

where $\mathbf{M}'$ is an adjusted attention map, Softmax$(\cdot)$ is the softmax activation function, and Topk$(\cdot)$ is the TopK sparse operation. Compared to $\mathbf{M}$, $\mathbf{M}'$ sheds light on more related elements and neglects those irrelevant. Thus, the neglected elements are not considered, while the weights of the relevant ones can be enlarged. Based on the TopK sparse operation, we perform two separate TopK operations with different values of k (represented as k_1 and k_2) on each $\mathbf{M}$. Subsequently, two learnable parameters are introduced to automatically adjust the ratio of these two TopK operations, enabling the model to capture diverse feature information during the fusion stage and enhancing the model's perceptual capability. To ensure that MSC can automatically select them, (5) can be rewritten as follows:

$$\mathbf{M}' = \alpha \cdot \text{Softmax}\big(\text{Topk}_1(\mathbf{M})\big) + \beta \cdot \text{Softmax}\big(\text{Topk}_2(\mathbf{M})\big) \tag{6}$$

where α and β are learnable parameters, and $\text{Topk}_1(\cdot)$ and $\text{Topk}_2(\cdot)$ are the functions of the TopK sparse operations with k_1 and k_2. Finally, the adjusted attention map $\mathbf{M}'$ is assigned to $\mathbf{V}$ by

$$\mathbf{A} = \mathbf{M}' \times \mathbf{V} \tag{7}$$

where $\mathbf{A} \in \mathbb{R}^{N_1 \times d}$ is the output of MSC, which integrated information from the two inputs.

Based on MSC, MFCM fuses $\{\mathbf{F}'_1, \mathbf{F}'_2, \mathbf{F}'_3\}$ with $\mathbf{F}'_4$ one by one, and then obtains the classification feature $\mathbf{F} \in \mathbb{R}^{H_4 \times W_4 \times 4C_2}$ through a simple concatenation. This step can be formulated as follows:

$$\begin{aligned} \mathbf{A}_i &= \text{MSC}\big(\mathbf{F}'_4, \mathbf{F}'_i\big), i = 1, 2, 3 \\ \mathbf{F} &= \text{Concat}\big(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{F}'_4\big) \end{aligned} \tag{8}$$

where MSC$(\cdot)$ means the function of MSC, and Concat$(\cdot)$ indicates the concatenation. Here, the reason for gradual fusion is that $\mathbf{F}'_4$ can extract information from various low-level features, making it more comprehensive and discriminative.

C. Group CBAM Enhancer

Although $\mathbf{F}$ can be used to complete the classification directly, it still includes redundant clues that limit its discrimination. Thus, we propose GCE to enhance $\mathbf{F}$ from the local and global aspects simultaneously. It is primarily based on CBAM, but it enhances the extraction of local salient features by introducing a grouping strategy. Furthermore, it reallocates weights based on this, improving the focus on critical regions. The architecture of GCE is illustrated in Fig. 6. For the input $\mathbf{F}$, we first uniformly split it into g groups $\{\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_g\}$ along the channel dimension. Then, independent CBAMs are applied to them respectively to get the local attention weights

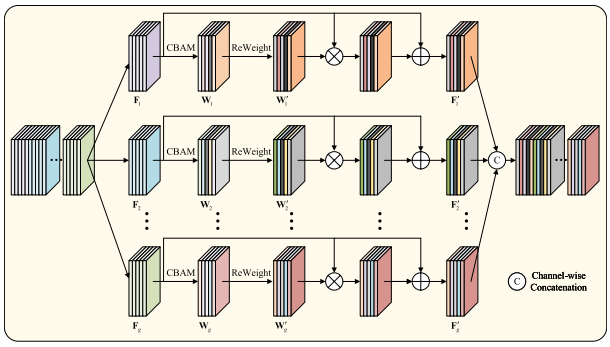


Fig. 6. Flowchart of the proposed GCE, which applies CBAM in a grouped manner to identify local salient information. Additionally, it readjusts the CBAM weights to highlight critical information further.

$$\mathbf{W}_i = \text{CBAM}(\mathbf{F}_i), i = 1, 2, \ldots, g \tag{9}$$

where CBAM$(\cdot)$ implies the function of CBAM, and $\mathbf{W}_i \in \mathbb{R}^{H_4 \times W_4 \times (4C_2/g)}$ is the CBAM weights for the ith part $\mathbf{F}_i$. The values in $\mathbf{W}_i$ range from 0 to 1, and a higher value means the corresponding information is more important. When we get $\mathbf{W}_i$, $\mathbf{F}_i$ can be reweighted by multiplying it. Here, to highlight the key clues in RS images, we readjust the CBAM weights. Specifically, we first calculate the mean of each weight $\mathbf{W}_i$. Then, the elements that are higher than the mean value are set to be 1. For the elements that are lower than the mean value, their values remain. After this process, the upgraded weights $\{\mathbf{W}'_1, \ldots, \mathbf{W}'_g\}$ can be obtained. Subsequently, we multiply them with $\{\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_g\}$, followed by a residual connection. This step can be defined as follows:

$$\mathbf{F}'_i = \mathbf{F}_i \cdot \mathbf{W}'_i + \mathbf{F}_i, i = 1, 2, \ldots, g. \tag{10}$$

Finally, we concatenate them along the channel dimension to produce the enhanced feature $\hat{F} \in \mathbb{R}^{H_4 \times W_4 \times 4C_2}$.

D. Classification Module

When we get $\hat{F}$, a global average pooling operation is applied to reduce its dimension. Then, a fully connected layer is employed to produce the classification outputs $\mathbf{F}_{\text{final}} \in \mathbb{R}^C$, where C represents the number of classification categories. Next, we select the cross-entropy loss function [56] to optimize our network. It can be defined as follows:

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log\big(p_{i,c}\big) \tag{11}$$

where N represents the number of samples, C denotes the number of classification categories, $y_{i,c}$ represents the ground-truth category of the ith sample as c, while $p_{i,c}$ represents the probability that the ith sample is predicted as the cth category.

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Fig. 7. Examples of the UCM dataset. The corresponding categories are also provided for convenience.



Fig. 8. Examples of the AID dataset. The corresponding categories are also provided for convenience.

IV. Experiment

A. Dataset Introduction

This article selects three public datasets to study the usefulness of MSCN.

1) UCM Land Use Dataset: The UC Merced (UCM) dataset [18] was constructed and released by the UC Merced computer vision laboratory. It encompasses 21 categories, such as "agricultural," "dense residential," and "storage tanks." There are 100 RS scenes in each category. Their spatial sizes and pixel resolutions are 256 × 256 and 1 foot. Examples of the UCM dataset are depicted in Fig. 7. To comprehensively evaluate our model's performance, we randomly select 50% and 80% RS scenes from UCM to establish the training sets, and the remaining RS scenes are regarded as the testing sets.

2) Aerial Image Dataset: The Aerial Image Dataset (AID) [57] was constructed jointly by Wuhan University and Huazhong University of Science and Technology in 2017. It comprises 30 classes, each of which contains approximately 220 to 420 RS scenes, resulting in a total of 10 000 scenes. Their spatial sizes are 600 × 600, and their spatial resolutions range from 0.5 to 8 m. Examples and the corresponding class names from the AID dataset are illustrated in Fig. 8. To assess the proposed model completely, we create two training sets, which randomly contain 20% and 50% scenes. The remaining scenes are deemed as the testing data.

3) Northwestern Polytechnical University-RS Image Scene Classification 45 (NWPU) Dataset: The NWPU dataset [38] was introduced by Northwestern Polytechnical University. There are 31 500 RS scenes in NWPU, which are divided into 45 classes equally. The RS scenes have 256 × 256 pixels, and their pixel resolutions range from 30 to 0.2 m. Examples of the NWPU dataset are illustrated in Fig. 9. NWPU presents a greater challenge for accurate classification due to its large volume and complex classes compared to the UCM and AID datasets. For model performance estimation, we randomly choose 10% and 20% scenes to establish the training sets, and the rest of the scenes are used to count the results.

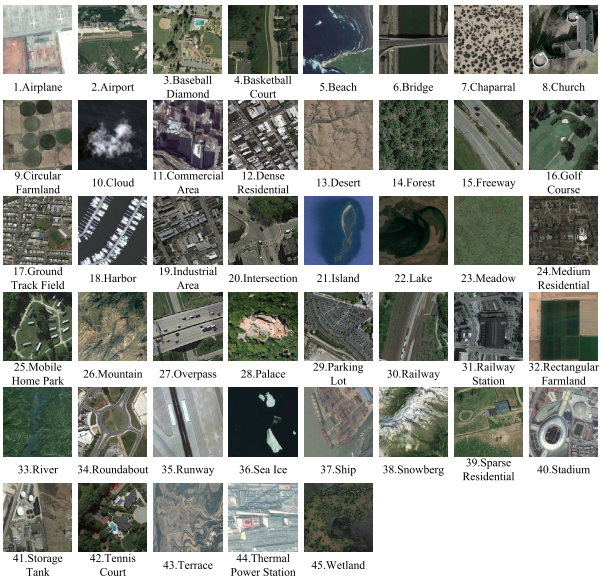


Fig. 9. Examples of the NWPU dataset. The corresponding categories are also provided for convenience.

B. Experimental Settings

All experiments are conducted on a workstation equipped with a GeForce RTX 3090 GPU, using the PyTorch platform with a version of 1.13.1. ResNet50 of MSCN is initialized using the pretrained parameters from ImageNet [58], and the other parts are initialized randomly. We select AdamW [59] as the optimizer to train the proposed MSCN. The epochs and batch size are set to be 200 and 128, respectively. The first five epochs are warm-up processes, and their learning rate equals $1e-6$. For other epochs, the initial learning rate is $1e-4$, and a cosine decay strategy is applied to adjust the learning rate automatically. The minimum learning rate is set as $1e-6$. To ensure the stability of the training stage, we adopt the same data augmentation methods as described in [47], including rand augmentation [60] and mixup [61]. All RS scenes are resized to 224 × 224 before feeding them into our model. For MSC, we adopt the multihead strategy [30] to better capture the relationship between different input features, and the number of heads is set to eight. Overall accuracy (OA) [56] is used as the evaluation metric, which represents the proportion of correctly classified scenes among the total images in the dataset. We repeat each experiment five times and report the mean and standard deviation (STD) values. Additionally, we plot confusion matrices [56] to visualize category-specific performance.

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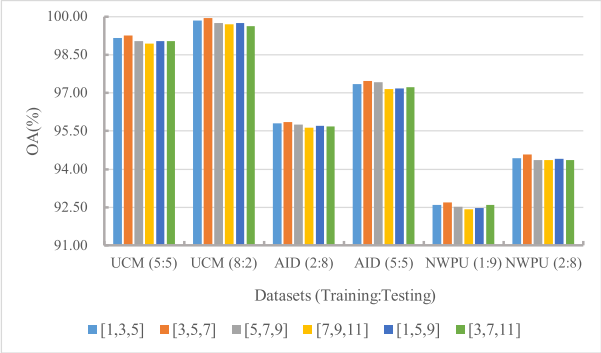


Fig. 10. OA values of different MSCNs with various pooling window sizes counted on the UCM, AID, and NWPU datasets.

C. Parameters Analysis

Several free parameters in our MSCN should be set in advance, they are the pooling window sizes in MSC, the threshold k_1 and k_2 in the TopK operation, and the group number g in GCE. In this section, we aim to empirically find their optical values for the three datasets.

1) Analysis of Pooling Window Sizes in MSC: As mentioned in Section III-B, MSC applies three average pooling operations with different window sizes to the input data to explore the multiscale clues. To find a proper set of pooling window sizes for different datasets, we vary the window sizes from 1 to 11 and make four combinations, i.e., $[1, 3, 5]$, $[3, 5, 7]$, $[5, 7, 9]$, and $[7, 9, 11]$. Additionally, to perform a more comprehensive analysis of the impact of pooling windows, we select larger window intervals, i.e., $[1, 5, 9]$ and $[3, 7, 11]$. Then, different MSCNs are trained with them by using different RS scenes, and their classification results are summarized in Fig. 10.

By observing the bars, we find that MSCN achieves the best performance for different datasets when the pooling window sizes are $[3, 5, 7]$. This is because the window size of 3 corresponds to a smaller receptive field, aiding in capturing local details. On the other hand, the window size of 7 corresponds to a larger receptive field, facilitating the capture of a broader range of overall information. While the window size of 5, being in the middle, helps to balance the capture of local and overall information. Combining these three values enables a more comprehensive exploration of multiscale information in RS images. Relatively, the smaller or larger pooling windows would result in information loss and a decline in classification performance. Moreover, as the window intervals increase (such as $[1, 5, 9]$ and $[3, 7, 11]$), the performance of MSCN does not improve. This is because larger intervals result in greater differences between the pooled feature maps, leading to less effective feature fusion. Based on the above analysis, we select $[3, 5, 7]$ as the optimal pooling window sizes.

2) Analysis of Threshold k_1 and k_2 in the TopK Operation: We apply the TopK sparse operation in MSC to fuse useful information from shallow to deep features. There are two parameters to control the fusion results, i.e., k_1 and k_2. We tune their values simultaneously to determine their optimal values. Considering that the functions of k_1 and k_2 are to remove the interference information and retain enough valuable clues, we set the ranges of k_1 and k_2 as $[(1/2), (1/4), (1/8), (1/16)]$ and $[(1/3), (1/6), (1/9)]$. The classification results of different MSCNs with various k_1 and k_2 across the three datasets are shown in Fig. 11.

We can observe that classification accuracies decrease for the UCM and AID datasets as the value of k_1 decreases. When k_2 is greater than $(1/6)$, the experimental results increase as the value of k_2 decreases. Conversely, when the value of k_2 continuously decreases, the accuracies would decline. For the NWPU dataset, when the values of k_1 and k_2 are greater than $(1/8)$ and $(1/6)$, respectively, the experimental results increase as their values decrease. However, classification accuracies decrease as their values continuously decrease. This is because lower values of k_1 and k_2 make MSCN exclude more interference information so that the results can be improved. Nevertheless, once their values are lower than those of $[(1/2), (1/6)]$ for UCM and AID datasets and $[(1/8), (1/6)]$ for NWPU dataset, too many messages, including some important details, will be eliminated. Thus, the classification results will get worse and worse. Finally, as suggested by Fig. 11, the optimal k_1 and k_2 values for the UCM, AID, and NWPU datasets are $[(1/2), (1/6)]$, $[(1/2), (1/6)]$, and $[(1/8), (1/6)]$, respectively.

To visually demonstrate the effectiveness of MSC in reducing the influence of interference areas during the feature fusion stage, we visualize the information discarded and retained by $\mathbf{F}'_4$ and $\mathbf{F}'_3$ at different k values, as shown in Fig. 12. The red dot area in the figure represents the query information, the semitransparent masked regions indicate the discarded information, and the remaining information is retained for feature fusion. The results indicate that a larger k_1 value retains more information, whereas a smaller k_2 value discards more irrelevant information while preserving more relevant data. Additionally, the retained information varies depending on the query. For example, when the sample is "Bridge," the query is related to the bridge, and more information related to the bridge is retained. In contrast, more information relevant to the water body can be preserved when the query is about the water body.

3) Analysis of g in GCE: In GCE, we split the input feature into g parts along the channel dimension, and then the local CBAM operation is applied to different parts to get the diverse attention weights. This way, the critical information hidden in the input data can be highlighted accordingly. To examine the impact of g on our model, we change its value at a range of $[1, 2, 4, 8, 16]$. The classification results of MSCNs trained by different datasets under varied g are displayed in Fig. 13. Note that when g equals 1, GCE is a global CBAM enhancer. From the observation of the bars, it is easy to find that the performance of MSCNs rises when g varies from 1 to 8. This indicates that the group partition strategy is useful in exploring the essential clues. However, the behavior of MSCNs weakens when g increases to 16. The main reason is that when g becomes too large, the originally correlated channel information is disrupted, leading to poor classification results. Another interesting observation is that when using the AID dataset and the ratio of training to testing samples is 5:5, the impact of different values of g on the model is negligible. This is because the AID dataset has a large enough sample size.

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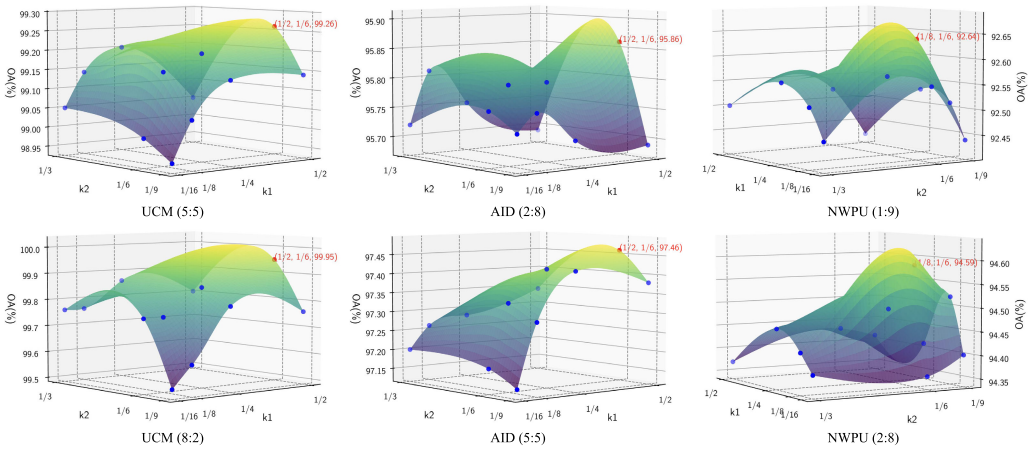


Fig. 11. OA values of MSCN with different k_1 and k_2 in MSC counted on the UCM, AID, and NWPU datasets. We fit the experimental results into a 3-D graph, where the data points corresponding to different values of k_1 and k_2 are indicated by blue dots. The maximum value is highlighted with a red dot, and its coordinates are annotated.

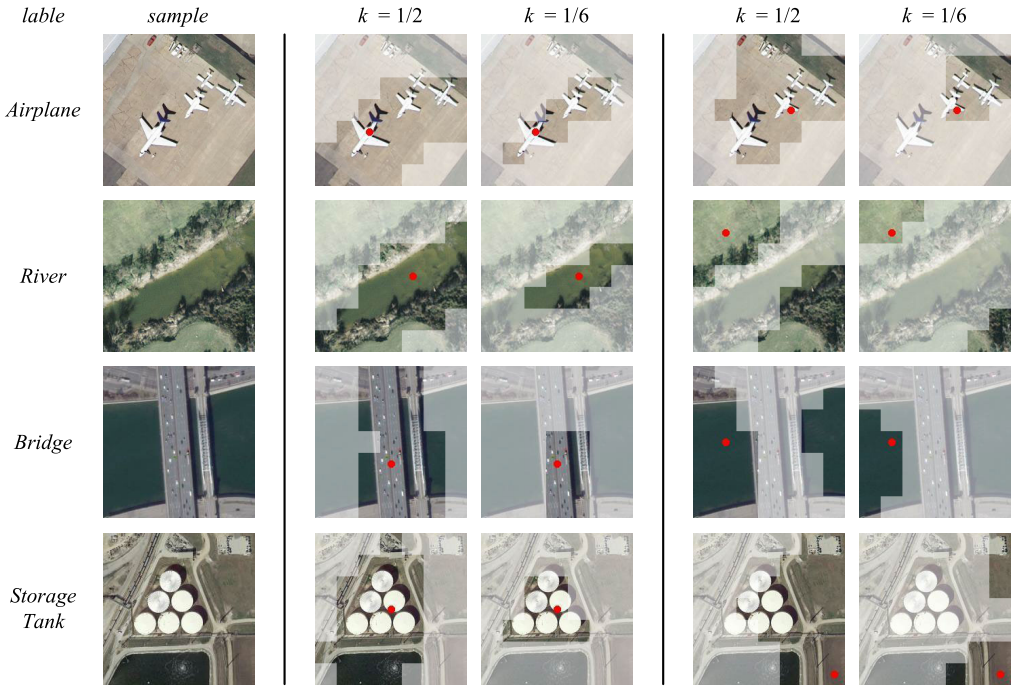


Fig. 12. Visualization results of TopK in MSC. The red dot area in the figure represents the query information, the semitransparent masked regions indicate the discarded information, and the remaining information is retained for feature fusion.

With equal ratios of training and testing samples, the model can learn features effectively during training and evaluate its generalization ability adequately during testing. Thus, feature enhancement has a minimal impact on the model. Overall, we suggest the optimal value of g is eight.

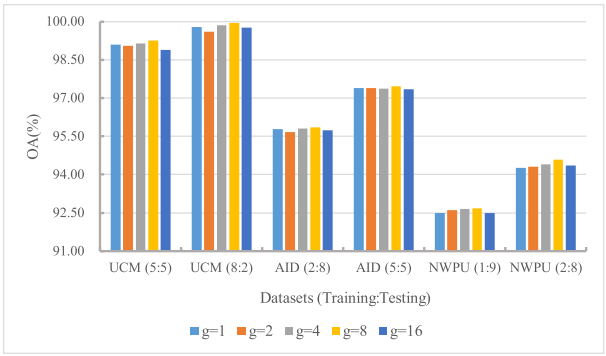


Fig. 13. OA values of different MSCNs with various g values counted on the UCM, AID, and NWPU datasets.

D. Comparisons of State-of-the-Art Models

To evaluate our proposed MSCN, we compare it with fifteen existing methods, and their detailed information can be found in Table I. Here, CrossViT-15-D [47] and Swin-T [51] are general-purpose models, while others are proposed in the RS community. MF2CNet, EMTCAL, EFCOMFF-Net, and MGML [62] are multilevel feature fusion-based models. TCNN [63], AGOS(ResNet-50) [64], MSGNet [65], SAGN [66], HFAM [67], and Upetu [68] are deep feature-based models. SKAL [69] and CGINet [70] are global and local feature fusion-based models. The backbones

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TABLE I
DETAILED INFORMATION OF THE COMPARED METHODS

Methods	Source and Year
CrossViT-15-D [47]	ICCV 2021
Swin-T [51]	ICCV 2021
MF2CNet [71]	TGRS 2022
TCNN [63]	TGRS 2022
EMTCAL [48]	TGRS 2022
SKAL [69]	TNNLS 2022
AGOS(ResNet-50) [64]	TGRS 2022
MSGNet [65]	TGRS 2023
SAGN [66]	TIP 2023
EMSCNet [74]	TGRS 2023
EFCOMFF-Net [23]	TGRS 2023
MGML [62]	TNNLS 2023
HFAM [67]	TGRS 2024
CGINet [70]	TGRS 2024
UPetu [68]	TGRS 2024

TABLE II
OA VALUES AND STDS (%) OF VARIOUS APPROACHES COUNTED ON UCM DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods	50% Training	80% Training
CrossViT-15-D [47]	98.99±0.16	99.76±0.15
Swin-T [51]	99.14±0.10	99.81±0.18
MF2CNet [71]	98.76±0.18	99.52±0.25
TCNN [63]	98.80±0.17	99.33±0.11
EMTCAL [48]	98.67±0.16	99.57±0.28
SKAL [69]	98.15±0.05	99.52±0.24
AGOS(ResNet-50) [64]	99.24±0.20	99.71±0.25
MSGNet [65]	99.03±0.11	99.76±0.14
SAGN [66]	98.95±0.08	99.82±0.10
EMSCNet [74]	98.70±0.46	99.44±0.16
EFCOMFF-Net [23]	99.11±0.10	99.86±0.12
MGML [62]	99.07±0.17	99.81±0.18
HFAM [67]	98.35±0.14	99.43±0.13
CGINet [70]	99.14±0.16	99.84±0.16
UPetu [68]	98.95±0.12	99.05±0.28
MSCN (ours)	**99.26±0.07**	**99.95±0.10**

of MF2CNet [71] and UPetu [68] are octave convolution [72] and ConvNexXt [73], respectively, while others use ResNet [17]. All compared models are accomplished using the released codes by the authors. Their free parameters are set according to the original literature, and their experimental settings are consistent with the contents described in Section IV-B.

1) Results on UCM: The results of all the models counted on the UCM dataset are shown in Table II. All methods achieve commendable classification results. Two general-purpose models (especially Swin-T) perform well, which illustrates the effectiveness of the Transformer. Even without considering the characteristics of RS images, the Transformer still helps them achieve good classification results. However, the proposed MSCN outperforms them in all cases, demonstrating our model's superiorities. Among the multilevel feature fusion methods, EFCOMFF-Net and MGML demonstrate the most vital behavior, primarily since they reduce discrepancies between multilevel features during the fusion stage. Even so, our MSCN outperforms them, achieving OA improvements of 0.15% and 0.09% over EFCOMFF-Net and 0.19% and 0.14% over MGML at 50% and 80% training ratios, respectively. For the deep feature-based methods, AGOS (ResNet-50) performs the best, with results only 0.02% and 0.24% lower than our approach. The strongest global and local feature fusion-based model is CGINet. This is because its specific integration scheme leads to the obtained features that can comprehensively represent RS scenes. Even so, our MSCN's OA values are still 0.12% and 0.11% higher than CGINet's. Overall, MSCN outperforms other methods in all cases. These encouraging findings can be attributed to the following three points. First, MSCN fuses multilevel features step by step, which is beneficial for integrating the different clues thoroughly so that a comprehensive feature representation can be generated. Second, the MSC-based fusion strategy not only extracts multiscale information from RS scenes but also reduces the negative impacts of interfering knowledge. Finally, GCE further emphasizes the key messages within the classification features, leading to better predictions.

Furthermore, we present the confusion matrices achieved by MSCN in Fig. 14. Note that these two confusion matrices are counted by the best results among the five repeated experiments. Analysis of Fig. 14(a) reveals that MSCN correctly classifies almost all categories, with only several samples from "golf course" and "medium residential" scenes being incorrectly classified. When the training ratio is 80%, all classes are correctly classified, demonstrating the powerful classification ability of MSCN.

2) Results on AID Dataset: The results of different methods counted on the AID dataset are shown in Table III. Similarly, the OA values of our MSCN are the highest among all compared models in all cases. When the training ratio equals 20%, MSCN's OA value is 95.86%, while Swin-T, MF2CNet, MSGNet, EFCOMFF-Net, and MGML's OA values are 95.45%, 95.54%, 95.46%, 95.76%, and 95.47, respectively. Compared to other models, our advantages are clear, surpassing CrossViT-15-D by 0.55%, TCNN by 1.31%, EMTCAL by 1.17%, SKAL by 1.48%, AGOS (ResNet-50) by 0.87%, SAGN by 0.69%, HFAM by 1.15%, CGINet by 0.52%, and UPetu by 0.53%. When the training ratio comes to 50%, our model surpasses the second-best Swin-T and EFCOMFF-Net by over 0.22%. The performance of MSGNet is also good, which is only 0.28% less than ours. However, the performance gaps between MSCN and other models are distinct, surpassing CrossViT-15-D by 0.50%, MF2CNet by 0.44%, TCNN by 0.74%, EMTCAL by 1.05%, SKAL by 0.70%, AGOS (ResNet-50) by 0.45%, SAGN by 0.69%, EMSCNet by 0.50%, MGML by 0.41%, HFAM by 0.73%, CGINet by 0.36%, and UPetu by 0.42%. The AID dataset contains more scenes and a wider variety of categories than the UCM dataset, making it more challenging to classify accurately. The leading performance of MSCN discussed above demonstrates its strong stability and generalization capability.

The confusion matrices of MSCN for AID are depicted in Fig. 15. Whether the proportion of training samples is 20% or 50%, most samples from different categories can be classified accurately by our model. For instance, all scenes corresponding to "river" are correctly classified. However, we also find that MSCN performs weakly in the "pond" category, with most

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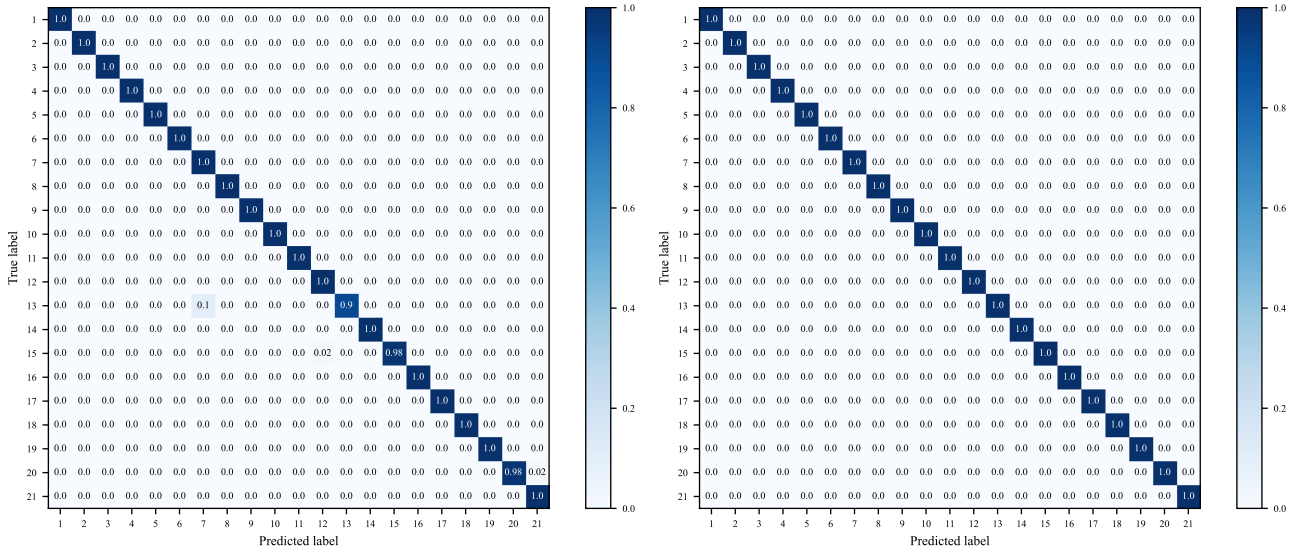


Fig. 14. Confusion matrices of MSCN counted on UCM dataset. The scene categories corresponding to the numbers are consistent with Fig. 7. (a) Confusion matrix for 50% training. (b) Confusion matrix for 80% training.

TABLE III

OA VALUES AND STDS (%) OF VARIOUS APPROACHES COUNTED ON AID DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods	20% training	50% training
CrossViT-15-D [47]	95.31±0.09	96.96±0.10
Swin-T [51]	95.45±0.13	97.24±0.13
MF2CNet [71]	95.54±0.17	97.02±0.28
TCNN [63]	94.55±0.27	96.72±0.23
EMTCAL [48]	94.69±0.14	96.41±0.23
SKAL [69]	94.38±0.10	96.76±0.20
AGOS(ResNet-50) [64]	94.99±0.24	97.01±0.18
MSGNet [65]	95.46±0.21	97.18±0.16
SAGN [66]	95.17±0.12	96.77±0.18
EMSCNet [74]	95.13±0.10	96.96±0.10
EFCOMFF-Net [23]	95.76±0.08	97.24±0.10
MGML [62]	95.47±0.06	97.05±0.10
HFAM [67]	94.71±0.06	96.73±0.04
CGINet [70]	95.34±0.14	97.10±0.24
UPetu [68]	95.33±0.10	97.04±0.21
MSCN(ours)	**95.86±0.16**	**97.46±0.12**

TABLE IV

OA VALUES AND STDS (%) OF VARIOUS APPROACHES COUNTED ON NWPU DATASET. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Methods	10% training	20% training
CrossViT-15-D [47]	91.95±0.15	94.03±0.13
Swin-T [51]	92.11±0.14	94.14±0.19
MF2CNet [71]	92.07±0.22	93.85±0.27
TCNN [63]	90.25±0.14	93.05±0.12
EMTCAL [48]	91.63±0.19	93.65±0.12
SKAL [69]	90.04±0.15	92.79±0.11
AGOS(ResNet-50) [64]	92.35±0.17	94.28±0.16
MSGNet [65]	92.40±0.16	94.57±0.12
SAGN [66]	91.73±0.18	93.49±0.10
EMSCNet [74]	92.16±0.07	94.08±0.20
EFCOMFF-Net [23]	92.44±0.08	94.46±0.09
MGML [62]	92.26±0.08	94.27±0.04
HFAM [67]	90.81±0.11	92.93±0.17
CGINet [70]	92.28±0.17	94.38±0.13
UPetu [68]	92.20±0.07	93.70±0.14
MSCN(ours)	**92.64±0.09**	**94.59±0.11**

images misclassified as "forest." This is because the water bodies in "pond" scenes often reflect the "forest" features, such as grass banks or shrub forests. Consequently, the network tends to misclassify them as "forest." This is an area where MSCN requires further improvement.

3) Results on NWPU Dataset: Compared to UCM and AID, the RS scenes in NWPU are large in volume and diverse in the category, enhancing the difficulties for accurate classification. The classification results of our MSCN and other compared models counted on NWPU are summarized in Table IV. The performance of MSCN maintains a leading position across various training ratios, demonstrating its powerful classification capabilities. When using 10% samples to train models, MSCN surpasses the second-best EFCOMFF-Net by 0.20%. Compared to other methods, the improvements achieved by MSCN are 0.69% (over CrossViT-15-D), 0.57% (over Swin-T), 0.57% (over MF2CNet), 2.39% (over TCNN), 1.01% (over EMTCAL), 2.60% (over SKAL), 0.29% (over AGOS (ResNet-50)), 0.24% (over MSGNet), 0.91% (over SAGN), 0.48% (over EMSCNet), 0.38% (over MGML), 1.83% (over HFAM), 0.36% (over CGINet), and 0.44% (over UPetu). When using 20% samples to train models, MSCN still performs best. It surpasses CrossViT-15-D by 0.56%, Swin-T by 0.45%, MF2CNet by 0.74%, TCNN by 1.54%, EMTCAL by 0.94%, SKAL by 1.80%, AGOS (ResNet-50) by 0.31%, SAGN by 1.10%, EMSCNet by 0.51%, MGML by 0.32%, HFAM by 3.78%, CGINet by 0.21%, and UPetu by 0.89%.

The confusion matrices of MSCN for NWPU are presented in Fig. 16. When the training ratio is 10% [see Fig. 16(a)], MSCN performs well in most categories. However, its behavior is weak in "tennis court" and "medium residential." Their samples are misclassified into each other's category. When the

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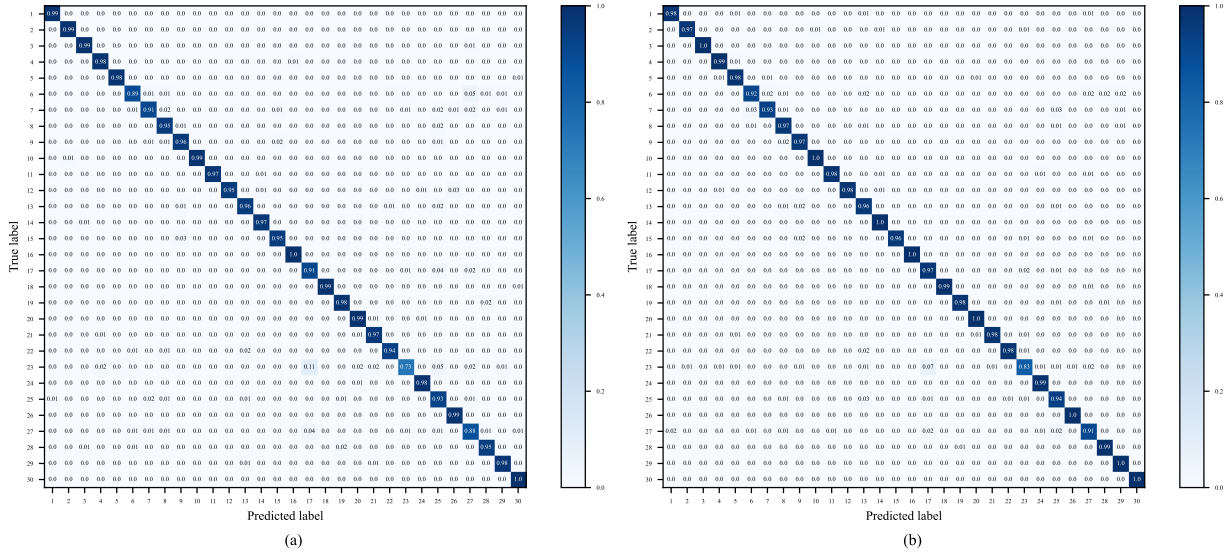


Fig. 15. Confusion matrices of MSCN counted on AID dataset. The scene categories corresponding to the numbers are consistent with Fig. 8. (a) Confusion matrix for 20% training. (b) Confusion matrix for 50% training.

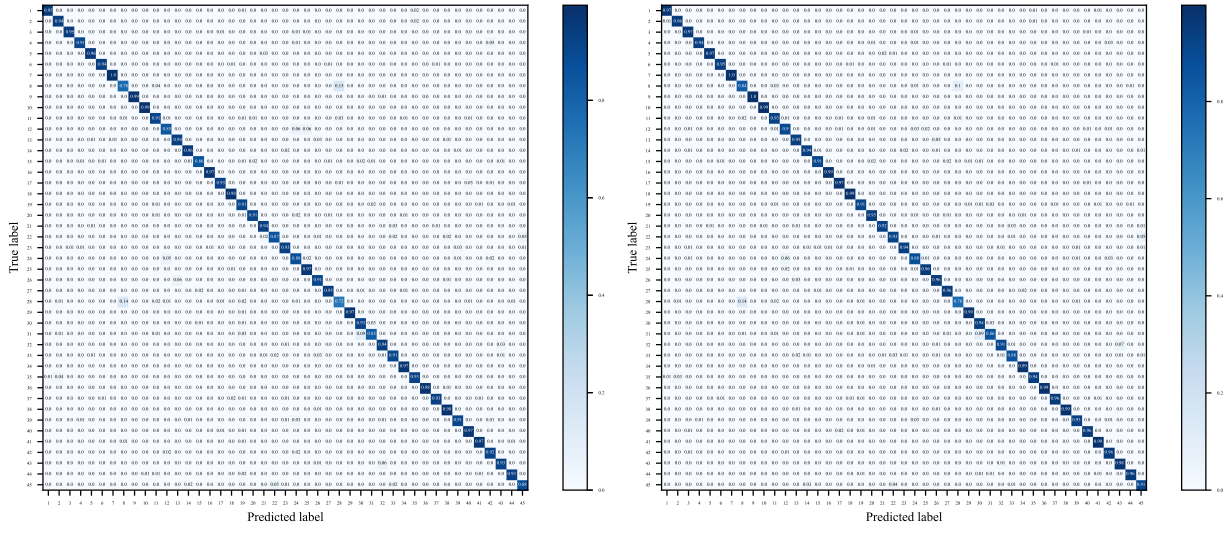


Fig. 16. Confusion matrices of MSCN counted on NWPU dataset. The scene categories corresponding to the numbers are consistent with Fig. 9. (a) Confusion matrix for 10% training. (b) Confusion matrix for 20% training.

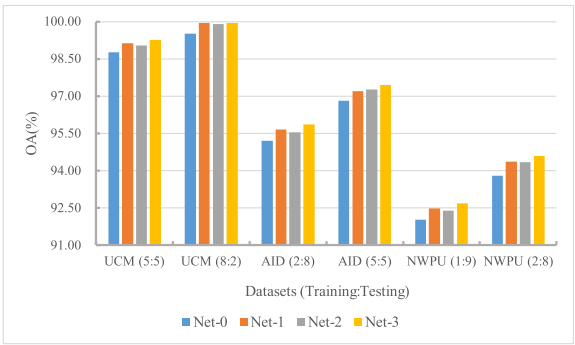


Fig. 17. OA values of Net-0 to Net-3 on the UCM, AID, and NWPU datasets.

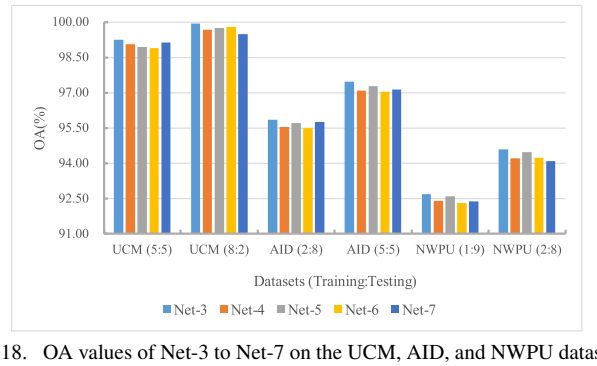


Fig. 18. OA values of Net-3 to Net-7 on the UCM, AID, and NWPU datasets.

training ratio rises to 20% [see Fig. 16(b)], the performance of different methods has improved. Although MSCN is still the best, it does not perform nicely for "tennis court" and "medium residential." The reasons behind this can be summarized as follows. Most "tennis court" scenes are surrounded by residential areas, resulting in high similarity between the two categories. In this scenario, our model is likely to produce incorrect classifications. Improving the model's ability to differentiate highly similar samples is one of our future tasks.

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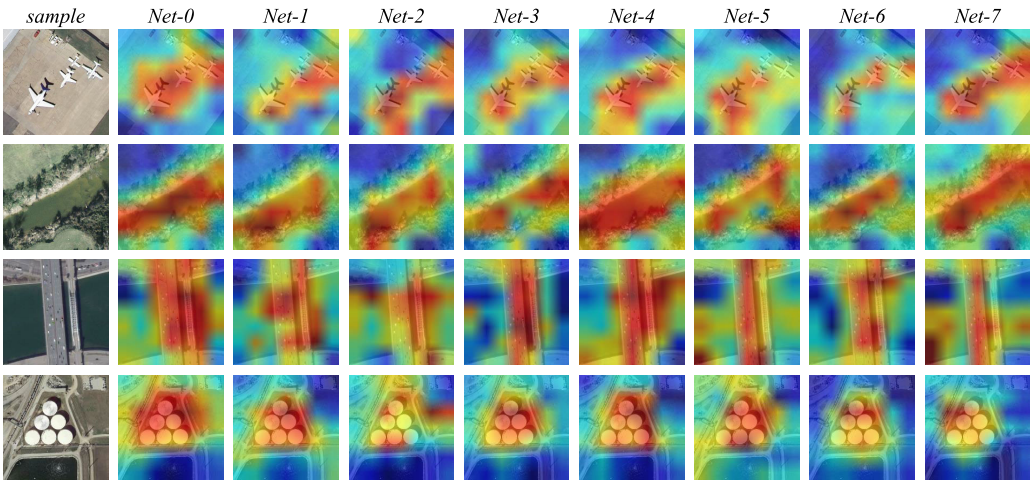


Fig. 19. Visualization results of Net-0 to Net-7. We select the classification features of each network and plot heat maps to obtain the visualization results.

E. Ablation Study

MFCM and GCE are two key parts of the proposed MSCN. MFCM is mainly composed of three MSCs, which are designed to extract valuable clues from multilevel features. GCE is an enhancer that effectively highlights key information within the classification feature. In this section, we plan to study their contributions. To this end, we construct four networks and analyze their performance on the three datasets. These four networks are as follows:

1) Net-0: MFEM + CM;
2) Net-1: MFEM + MFCM + CM;
3) Net-2: MFEM + GCE + CM;
4) Net-3: MFEM + MFCM + GCE + CM.

Net-0 is a baseline network where only MFEM is adopted as the feature extractor. To complete the classification, multilevel features learned by MFEM are concatenated and directly fed into CM. Net-1 is built upon Net-0 with the addition of MFCM. Instead of concatenating multilevel features, MFCM can fuse them effectively. Then, CM is also used to generate the predictions. Net-2 expands on Net-0 by incorporating GCE. In this model, the multilevel features are enhanced by GCE respectively. Then, the enhanced features are concatenated together. With the assistance of CE, the classification results can be obtained. Net-3 is actually the proposed MSCN. In Net-0 and Net-2, the 3×3 convolution in MFEM transforms features at each level to match the size of $\mathbf{F}'_4$ in Section III-A, ensuring consistent dimensions for concatenation.

The performance of Net-0 to Net-3 across the three datasets is depicted in Fig. 17. By observing those bars, the behavior of Net-1 and Net-2 is stronger than that of Net-0 in all cases, indicating that the feature fusion scheme in MFCM and the feature enhancement GCE play a positive role in our model. An interesting finding is that the OA values of Net-1 are higher than that of Net-2 in most cases. This suggests that MFCM has a more significant impact than GCE in our proposed MSCN. In other words, the multilevel feature fusion contributes more to the classification results. Also, when MFCM and GCE are combined, Net-3 outperforms Net-1 and Net-2. This denotes that these two modules can provide complementary clues for the final classification.

Additionally, to further demonstrate the effectiveness of our model, we designed the following ablation experiments, they are the following:

1) Net-4: MFEM + MFCM(CAM) + GCE + CM;
2) Net-5: MFEM + MFCM(only k_1 in MSC) + GCE + CM;
3) Net-6: MFEM + MFCM(only k_2 in MSC) + GCE + CM;
4) Net-7: MFEM + MFCM + CBAM + CM.

For Net-4, we replace all MSCs in MFCM with CAMs to evaluate the performance of MSC compared to CAM. For Net-5 and Net-6, we apply the TopK operation only with k_1 and k_2 individually for all MSCs to verify the necessity of selecting two different k values. For Net-7, instead of GCE, CBAM is adopted to assess the performance improvement brought by GCE over CBAM. The performance of Net-4 to Net-7 across the three datasets is depicted in Fig. 18. The behavior of Net-3 (i.e., MSCN) is also exhibited for reference.

From observing these bars, we can easily find that Net-3 outperforms other networks, which implies the importance of MSC, multiple k values, and GCE. Taking UCM with 50% training samples as an example, when we use CAM rather than MSC, Net-4's OA value is 0.21% lower than Net-3. This is because, compared to CAM, which only focuses on information fusion, MSC also pays attention to exploring the multiscale clues and removing the redundant messages. Thus, the fused results achieved by MSC can provide more helpful materials for subsequent operations. Taking AID with 20% training samples as another instance, once the multiple k-value scheme is exchanged with the single k-value design, the performance of Net-5 and Net-6 drops apparently compared with Net-3. This confirms the rationale behind our choice of k_1 and k_2 values. For Net-7, its behavior is weaker than that of Net-3 in all cases. This is mainly due to GCE's ability to extract key information in groups compared to CBAM, thereby improving the classification results.

Finally, we provided visualizations for each ablation experiment, as shown in Fig. 19. These heat maps reflect the focus

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areas of different models. Net-3 is the proposed MSCN. Taking the sample of "Airplane" as an example, the attention area of Net-3 precisely focuses on the airplanes' location in the image. In contrast, Net-0 has a more dispersed attention area, lacking specificity compared to Net-3. Net-1 shows insufficient focus on the airplane region due to the absence of GCE. Net-2 increases attention to irrelevant areas because it lacks MFCM. Net-4 employs CAM for feature fusion, but without filtering interference information, its overall performance is inferior to that of Net-3. Net-5 and Net-6 both utilize a single k value, which limits the breadth of their overall areas of focus. Additionally, because Net-6 employs a smaller k value, its area of focus is further narrowed compared to Net-5. While Net-7 effectively concentrates on target areas, it still experiences issues with excessive focus on some areas that are not targets.

V. Conclusion

This article presents a new RSSC network named MSCN. First, MSCN leverages a pretrained ResNet50 to extract multilevel features. The features are categorized into high-level and low-level groups based on their contributions. Then, the high-level features are cross-fused with different shallow features by the developed MFEM to extract useful information effectively. The proposed MFEM mainly consists of the MSC mechanism. By employing sparse operations, MSC can fuse multilevel features while filtering out irrelevant information, thus reducing the impact of interference messages. Finally, we introduce GCE to enhance the representation of classification features by identifying local salient information with the help of CBAM. Encouraging experimental results counted on multiple public RSSC datasets demonstrate that our MSCN is a helpful classification model. However, the network is prone to misclassification when the feature similarity between different scene categories is high or the scenes are closely related. Overcoming this deficiency is a key focus of our future work.

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Jingjing Ma (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2004 and 2012, respectively.

She is currently an Associate Professor at the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. Her research interests include computational intelligence and image understanding.

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Wei Jiang received the B.E. degree in automation from Yangzhou University, Yangzhou, China, in 2021. He is currently pursuing the M.E. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include machine learning and remote sensing image processing.



Xu Tang (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronic circuit and system from Xidian University, Xi'an, China, in 2007, 2010, and 2017, respectively.

From 2015 to 2016, he was a Joint Ph.D. Student along with Prof. W. J. Emery at the University of Colorado at Boulder, Boulder, CO, USA. He is currently a Full Professor at the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. His research interests include remote sensing image content-based retrieval and reranking, hyperspectral image processing, remote sensing scene classification, and object detection. For more details, please refer to: https://web.xidian.edu.cn/tangxu/en/index.html.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree from the School of Electronic Engineering, Xidian University, in 2006.

From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA. She is a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Fang Liu (Member, IEEE) was born in China, in 1990. She received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is currently an Associate Professor at Nanjing University of Science and Technology, Nanjing, China. Her research interests include deep learning, object detection, polarimetric SAR image classification, and change detection.



Licheng Jiao (Fellow, IEEE) received the B.S. degree in high voltage from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China. He has authored or co-authored more than 200 scientific articles. His research interests include signal and image processing, nonlinear circuits and systems theory, wavelet theory, natural computation, and intelligent information processing.

Dr. Jiao is a member of the IEEE Xian Section Executive Committee and an Executive Committee Member of the Chinese Association of Artificial Intelligence. He is the Chairperson of the Awards and Recognition Committee.

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